Symmetric Matrices

Hessian matrices.

为什么在优化问题中经常出现对称矩阵？

定义：Coindex.

A symmetric, The index of A Ind(A), is defined to be the number of negative eigenvalues (multiplicity counted).

The coindex of A, Coind(A), is the number of positive eigenvalues.

Hessian 矩阵：目标函数 $f(x)$ 的二阶偏导数矩阵 $\nabla^2 f(x)$，用于判断极值类型和收敛性.